AXIAL PISTON PUMPS / PRODUCT INFORMATION

P30 Axial Piston Pump

Compare the 23, 30 and 35 cc/rev (1.83 and 2.14 in³/rev) options against delivered flow and drive torque. Use the published curves at matching speed and pressure; separate maximum flow and pressure values do not establish one simultaneous operating point.

The P30 nearly doubles the maximum flow of the P15, stepping from 37.2 L/min (9.83 US gpm) to 66 L/min (17.4 US gpm) within the same oil-free, 316 stainless steel package. With three displacement options and the introduction of 1" BSPP ports on the unboosted variants, the P30 is designed for high-flow industrial applications where port velocity and system pressure drop matter.



WaterQuest P30 axial piston water hydraulic pump, 316 stainless steel construction

At a glance

Parameter / value	Parameter / value
Displacements 23/30/35 cc/rev (2.136 in³/rev)	Max Pressure 160 bar (2,321 psi) continuous
Max Flow 66 L/min (17.44 US gpm)	Max Input Power 19.5 kW (26.15 hp)
Shaft Key 8x7 mm	Body OD 150 mm ()

Selection basis

Match the selected configuration to the working water, pressure, flow, temperature and duty. Maximum ratings and separate curve points do not establish one simultaneous operating point. Confirm installation and supplied configuration with WaterQuest Solutions.

Technical data

Specifications and selection data

Parameter	23 cc/rev (1.4 in³/rev)	30 cc/rev (1.83 in³/rev)	35 cc/rev (2.14 in³/rev)	Note
Displacement	23 cc/rev (1.4 in³/rev)	30 cc/rev (1.83 in³/rev)	35 cc/rev (2.14 in³/rev)	Fixed
Max Continuous Pressure	160 bar (2,321 psi)			All configs
Max Flow (unboosted, 1800 rpm)	~39 L/min (10.3 US gpm)	~51 L/min (13.5 US gpm)	~59 L/min (15.6 US gpm)	PC / PA
Max Flow (boosted, 2000 rpm)	~44 L/min (11.6 US gpm)	~57 L/min (15.1 US gpm)	66 L/min (17.4 US gpm)	PB
Max RPM (Unboosted)	1800 rpm			PC / PA / PPC
Max RPM (Boosted)	2000 rpm			PB
Max Input Power	~12 kW (16.1 hp)	~16.5 kW (22.1 hp)	19.5 kW (26.1 hp)	At 1750 rpm, 160 bar (2,321 psi)
Max Torque at 160 bar (2,321 psi) (1500 rpm)	~64 Nm (47.2 lbf·ft)	~84 Nm (62 lbf·ft)	~94 Nm (69.3 lbf·ft)	All configs
Max Fluid Temperature	50°C (consult WQS above 50°C (122 °F))			
Min Fluid Temperature	2°C (35.6 °F) (antifreeze option available)			
Housing Material	316 Stainless Steel			
P Port - Unboosted (PC/PA/PPC)	1" BSPP			All displacements
T Port - Unboosted (PC/PA/PPC)	1" BSPP			All displacements
P Port - Boosted (PB)	3/4" BSPP (pressure outlet)			All displacements
T Port - Boosted (PB)	1/2" BSPP (case drain to sump)			Max 2.5 bar (36.3 psi) at T port
Shaft Diameter	23.993 / 23.980 mm (0.945 / 0.944 in)			
Shaft Key	8 x 7 mm (0.315 x 0.276 in)			
Flange / Spigot OD	101.60 / 101.55 mm (4 / 4 in)			
Body Outer Diameter	150 mm (5.91 in)			
Overall Body Length	156 mm (6.14 in)			
Shaft Section Length	51 mm (2.01 in) total (40 mm (1.57 in) inner zone)			
Key Projection	9.4 mm (0.37 in)			
Base Projection	18 mm (0.709 in)			
Port Face	110 mm (4.33 in) square, 32 mm (1.26 in) port spacing			

Parameter	23 cc/rev (1.4 in³/rev)	30 cc/rev (1.83 in³/rev)	35 cc/rev (2.14 in³/rev)	Note
PPC M8 Mounting Holes PCD	116 mm (4.57 in) (PPC option only)			With O-ring face seal
Shaft Rotation	PC = CW, PA = CCW / PPC = CW / PB = bi-directional			
Inlet Filtration Required	10 µm (394 µin) nominal, 25 µm (984 µin) absolute, beta-10 = 75			
Certifications	Confirm applicability and supplied documentation: ATEX II 2GD ck T6, ISO 9001 FM 87247, F4N			

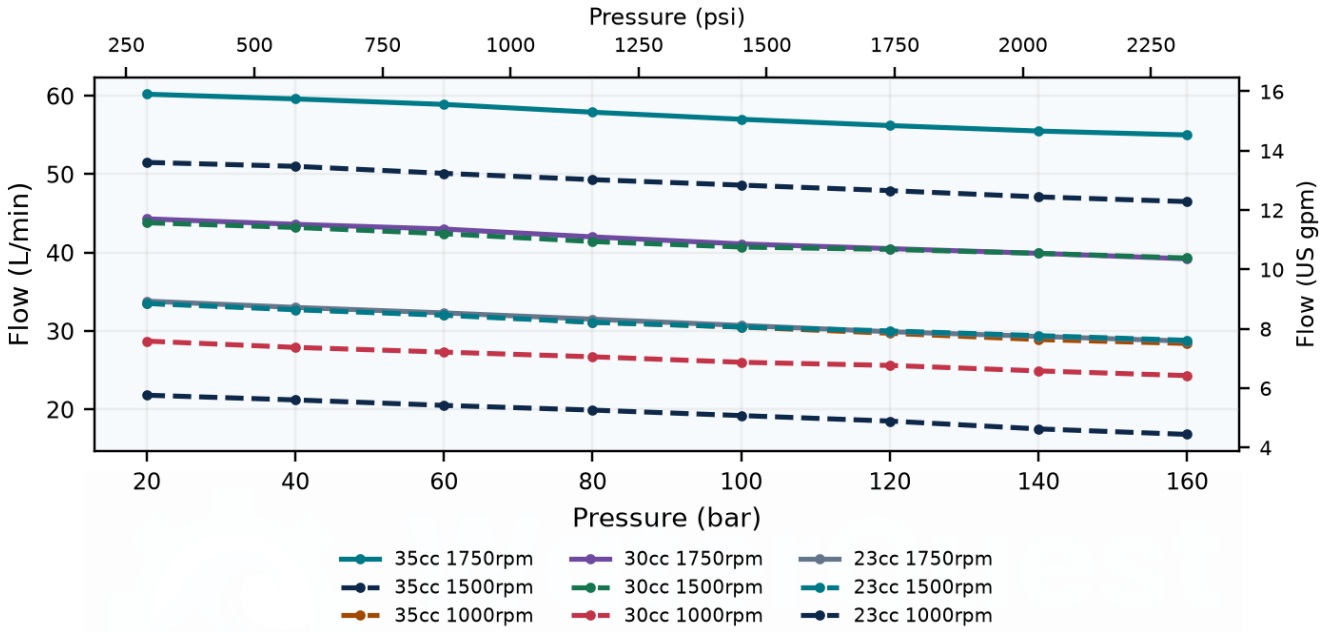
Selection and integration

All performance figures on clean tap water at 15°C (59 °F). The exact operating fluid must be specified on all enquiries and orders. 316 SS Material Test Reports available on request. Contact WaterQuest for motor frame sizing and bell housing selection across all three displacements.

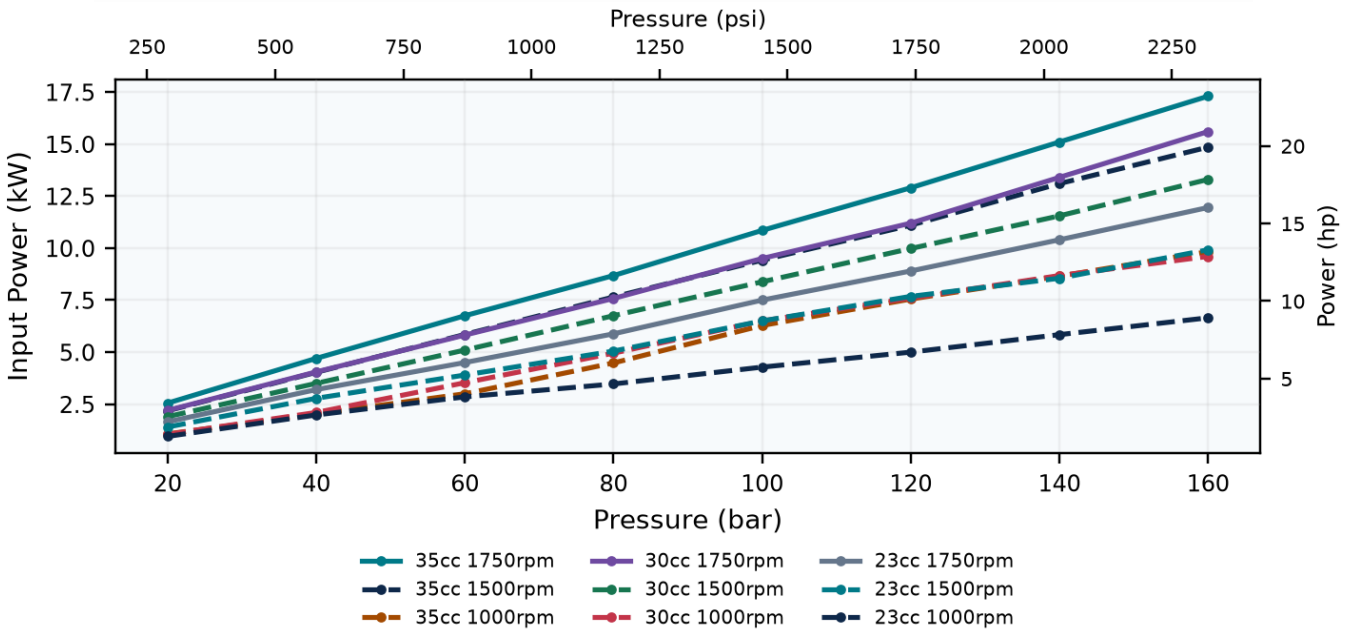
Performance curves

Reference curves reproduced from the product webpage. Curves retain the published data points and speed/displacement labels. Confirm guaranteed performance at the selected duty.

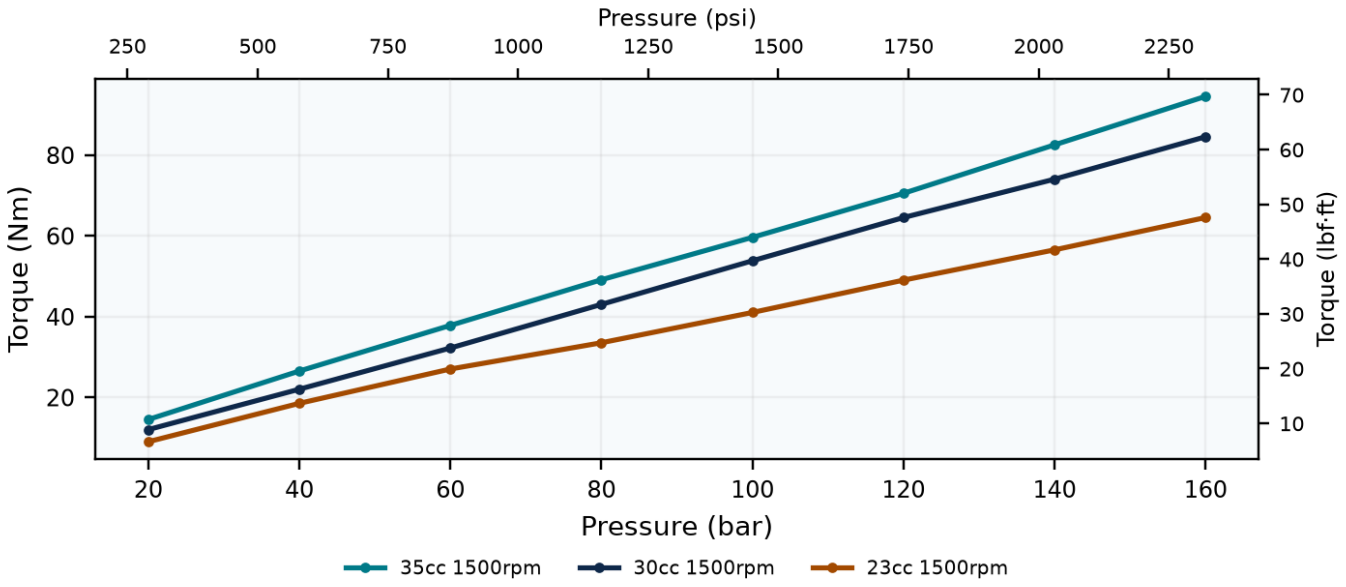
Flow (L/min) vs Pressure (bar)



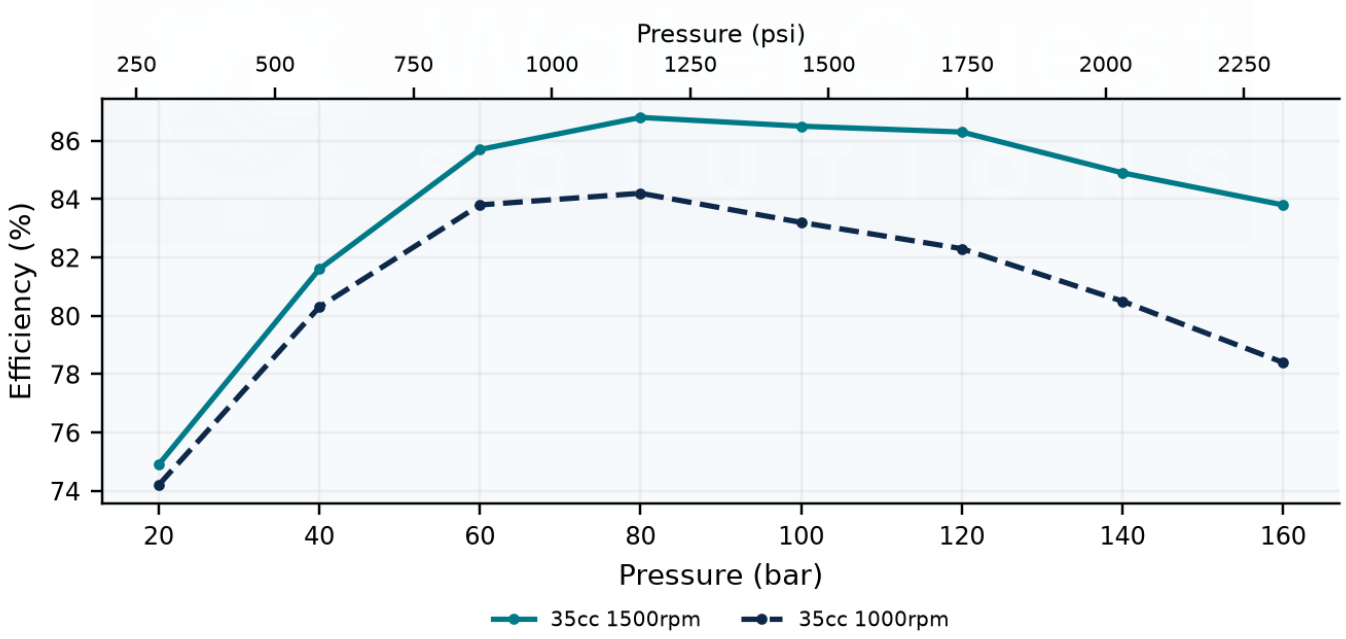
Input Power (kW) vs Pressure (bar)



Torque (Nm) vs Pressure (bar)



Efficiency (%) vs Pressure (bar)

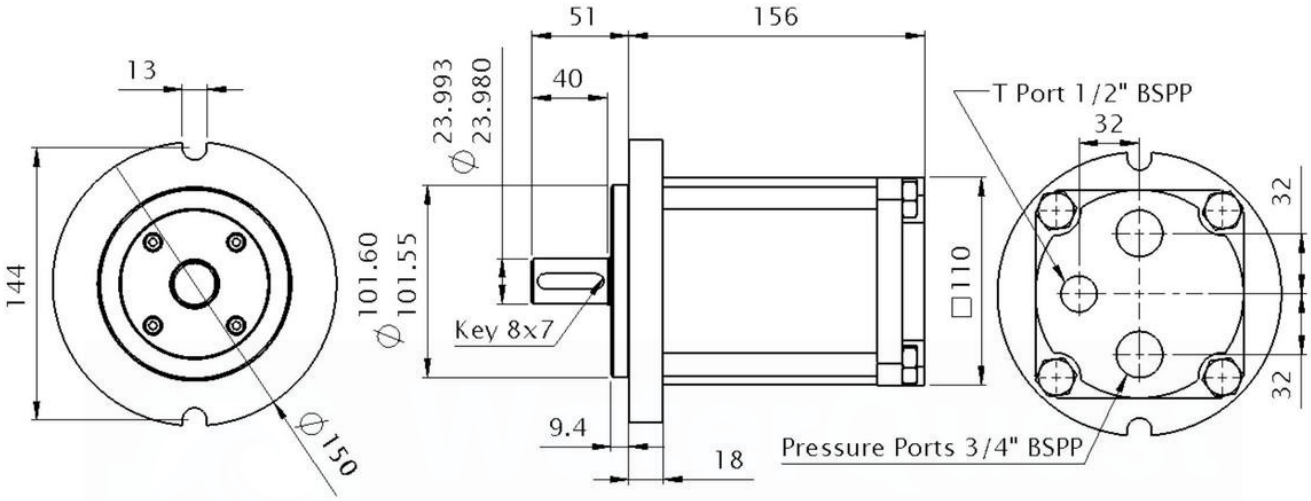


Drawings and reference visuals

Dimensions, ports and curves are reproduced from the webpage visuals. Use the original drawing units and confirm the current approved drawing before fabrication or installation.

PB160-23W, PB160-30W and PB160-35W boosted P30 dimensional drawing, 3/4 inch BSPP pressure ports, 1/2 inch BSPP T port

PB160-23W, PB160-30W & PB160-35W



PC160, PA160 and PPC160 unboosted P30 dimensional drawing, both ports 1 inch BSPP, PPC adds M8 holes on 116mm PCD

PC160-23W, PC160-30W & PC160-35W

PA160-23W, PA160-30W & PA160-35W

PPC160-23W, PPC160-30W & PPC160-35W (O ring face seal and M8 holes only added on PP Option)

